

Table 6.5. Panel Regression: Level Equation*(Dependent variable, NFA/GDP)*

Explanatory Variable	Fixed Effects	Ordinary Least Squares	Random Effects	Fixed Effects	Instrumental Variables
Government debt/GDP	-0.57** (0.05)	-0.28** (0.05)	-0.56** (0.04)	-0.41** (0.04)	-0.49* (0.23)
Dependency ratio	-0.45 (0.33)	-0.37* (0.20)	-0.40 (0.30)	-0.16 (0.30)	-1.29** (0.56)
Relative income	0.53** (0.20)	0.66** (0.08)	0.53** (0.17)	0.69* (0.18)	-0.72 (0.76)
Capital/output	—	—	—	-0.37** (0.04)	—
Constant	—	-38.4** (6.4)	-16.0 (12.1)	—	—
D.W.	0.36	0.26	—	—	—
$\bar{R}^2$	0.78	0.23	0.77	0.82	—
Panel unit root test [p-value]	0.14	0.01	—	0.44	—

Notes: *(**) denotes significance at the 10 (5) percent level; the standard errors are indicative only; number of countries (N) = 21; number of time periods (T) = 23; mean of dependent variable (NFA/GDP): -7.27; standard error of dependent variable: 32.48.

The explanatory variables in the levels specification are basically those from the cross-sectional regression estimated above, namely, the dependency ratio, the capital stock and relative income variables, and fiscal policy.⁴³ Corresponding to our use of the stock of net foreign assets as the dependent variable, we use the stock of government debt to represent fiscal effects. The results from estimating this levels regression are shown in Table 6.5.

The first column shows the estimates when country specific dummies (or fixed effects) are included. The second column shows the result of an OLS regression with a constant that is common across countries. The third column is the random effects estimate.⁴⁴ The slope coefficients are almost identical across the random and fixed effects specifications. The hypothesis that all the constant terms are the same is, however, rejected (critical value 60.1). The model estimated using the first difference estimator (shown in column 5) suggests that the fixed effect specification is still an appropriate specification.

The results again show a much larger impact of fiscal policy in this specification compared with the cross-sectional regression earlier. An increase in government debt of 1 percentage point of GDP is associated with a reduction in the NFA-to-GDP ratio of about 0.6 of a percentage point,⁴⁵ in line with the long-run impact of fiscal policy in the partial adjustment equation. The relationship between the level of government debt and the net foreign asset position is stronger possibly because the government debt variable may be better capturing the long-run impact of fiscal policy.

Demographics again have a negative although not significant impact on the NFA ratio in this specification.⁴⁶ Relative income has a positive influence on the current account, while the capital-output ratio has a negative influence in the fixed effects regressions, leading us to prefer column 1 over column 4. Again, the latter results may be caused either by the Scandinavian countries or by the collinearity between the capital-output ratio and relative income.

⁴³As the time series properties of the squared variables may be integrated are not obvious, we estimate the levels equation including the income and capital stock variables in linear form.

⁴⁴A Hausman test (with critical value 3.46) fails to reject the random effects model in favor of the fixed effects model.

⁴⁵Masson, Kremers, and Horne (1994) find coefficients of 0.3, 0.5, and 0.7, respectively, in Germany, Japan, and the United States for the effect of government debt on NFA.

⁴⁶Under a cointegration interpretation, note that the t -statistics from the levels regression have nonstandard limiting distributions and, hence, are only indicative.

Table 6.6. Panel Regression: Error-Correction Model(Dependent variable, Δ NFA/GDP)

Explanatory Variable	Fixed Effects	Ordinary Least Squares	Random Effects	Fixed Effects
Lagged change in NFA/GDP	0.40** (0.04)	0.44** (0.04)	0.33** (0.04)	0.40** (0.04)
Change in government debt/GDP	-0.47** (0.06)	-0.38** (0.06)	-0.36** (0.06)	-0.50** (0.06)
Real exchange rate change	-4.81 (4.81)	-3.23 (4.68)	2.80 (4.58)	0.20 (4.55)
First lag	2.45 (4.60)	3.23 (4.59)	1.01 (5.03)	1.48 (4.64)
Second lag	-8.65* (4.58)	-7.10 (4.51)	-6.99 (4.65)	-9.89** (4.60)
Domestic output gap	-0.45** (0.10)	-0.36** (0.09)	-0.18** (0.09)	-0.52** (0.10)
Error correction model term	-0.10** (0.02)	-0.10** (0.02)	-0.11** (0.02)	-0.10** (0.02)
Terms of trade change	0.13** (0.04)	0.13** (0.04)	0.07** (0.04)	—
Constant	—	-0.13 (0.25)	0.03 (0.26)	—
$LM[\chi^2(1)]$	3.2*	1.3	1.4	1.9
$LM[\chi^2(4)]$	32.4**	7.3	11.8**	28.7**
$\bar{R}^2$	0.40	0.40	0.37	0.39

Notes: * (**) denotes significance at the 10 (5) percent level; number of countries (N) = 21; number of time periods (T) = 20; mean of dependent variable: -0.98; standard error of dependent variable: 5.70.

The panel unit root test rejects the null of nonstationary residuals (i.e., no cointegration) in the OLS specification but narrowly fails to reject the null in our preferred fixed effects specification. The country dummies are jointly significant in the fixed effects specification. The hypothesis that the coefficients on the explanatory variables are the same across countries is again rejected, however (the F statistic is 28.4). Nevertheless, as stated above, the regression can be regarded as capturing the effect on an average country's net foreign asset position of changes in the explanatory variables.

The estimation results of the error-correction model are shown in Table 6.6. The short-run, error-correction specification includes the residuals (i.e., error-correction mechanism terms) from the levels regression estimated with fixed effects in column 1 of Table 6.5. The specification also includes other variables that may have shorter-term effects on the current account discussed earlier. The dependent variable in the short-run model is the change in the

net foreign assets ratio.⁴⁷ The presence of the lagged dependent variable in the regression suggests that fixed effects is the appropriate specification; moreover, the Hausman test rejects the random effects assumption at the 5 percent level.

The error-correction term was significant and had the expected negative sign. The coefficient of -0.1 in column 1 suggests that deviations from the equilibrium level of NFA to GDP have a half life of six or seven years. The other short-run variables generally have the same impact as in the current account estimation in Table 6.4. The exchange rate again has a J-

⁴⁷While the first difference of the ratio of net foreign assets to GDP used in the levels regression is not precisely the current-account-to-GDP ratio, as discussed earlier, the current-account-to-GDP ratio (ca) maps into the change in the NFA-to-GDP ratio (Δnfa) through the growth rate of output (g) in steady state. Similarly, the budget surplus maps into changes in the level of government debt.

curve type impact on the change in NFA when terms of trade are excluded in column 4, with the initial effect being positive before turning negative in the second year. This may in part reflect valuation effects on the stock of NFA. Fiscal policy, meanwhile, has a large influence on the change in NFA in the short run. A change in the stock of government debt of 1 percent of GDP will improve the NFA position in the short run by around 0.5 percent of GDP. The domestic output gap term enters significantly with the expected negative sign.

The error-correction model for NFA thus unites, in a general sense, the findings from the cross-sectional regressions (Table 6.2) and the panel estimates of the partial adjustment model (Table 6.4). In the long run, the stage of development, government debt, and demographics affect the current account, associated with longer-term stock equilibrium, but in the short run, the flow current account is affected by movements in relative prices, the stage of the business cycle, and the fiscal budget balance.

Partial Adjustment Model Extensions and the Saving-Investment Norm

The previous discussion has shown that, for our panel of 21 industrial countries, both a partial adjustment model of the current account and an error-correction model of net foreign assets can provide satisfactory explanations of the longer-term behavior of the current account in terms of explanatory variables that can be thought of as operating on the external balance through their effects on underlying saving and investment behavior. This section further examines—in the context of the partial adjustment model—the role of relative fiscal, demographic, and stage of development variables in determining the saving-investment balance; it does so to construct measures of “normal” current account positions (i.e., saving-investment norms) for our sample of industrial countries. The decision to base the saving-investment norms on the partial adjustment model, rather than the error-correction model of NFA, reflected an initial desire to have separate norms for saving and investment, along with the fact that adequate data were not available on cumulative stocks of saving and investment. In extending the partial adjustment model for constructing norms, we examine issues such as country-specific constants and coefficients, omitted variables, and the implications of the saving-investment model for the global discrepancy.

Following the reduced-form framework described in Section IV, we can express the current account as part of a larger system involving *separate* equations for saving, investment, and the saving-investment

balance (with the appropriate cross-equation restrictions). With common coefficients across countries, and expressing the set of explanatory variables in relative terms (as deviations from global averages), this framework has the desirable property of preserving the level of the global current account (i.e., global discrepancy) in the face of changes in own-country variables or their global averages—which can be thought of as reflecting changes in the world real interest rate. We will return to this system of equations shortly.

Partial Adjustment Model: Single Equation Estimates

Before turning to the system estimates, it is useful to describe many of the issues relevant for constructing the saving-investment norms in the context of a single equation for the saving-investment balance. Single-equation estimation of the partial adjustment model for our panel of 21 industrial countries over the sample period from 1972 to 1993 yields the following result:

$$\begin{aligned} CA = & 0.66 CA_{-1} + 0.15 SUR - 0.04 DEM \\ & + 0.05 YPCAP - 0.25 GAP + \epsilon; \\ \bar{R}^2 = & 0.76, D.W. = 1.76, S.E.E. = 1.81. \end{aligned} \quad (6.4)$$

In this equation, note that the current account, *CA*, is expressed as a ratio to GDP, but the fiscal surplus (as a share of GDP), *SUR*, and the dependency ratio, *DEM*, are now standardized, expressed as deviations from their GDP-weighted averages for the industrial countries; *YPCAP* is income per capita relative to the United States, and *GAP* is the domestic output gap.⁴⁸ The fitted equation also contains country-specific constants and a measure for German unification (1990–93).⁴⁹

⁴⁸A standardized measure for the stage of development variable, *YPCAP*, expressed as a deviation from its average, is considered later. The discussion in Section IV also suggests including a global average for the output gap (to standardize this variable). A foreign output gap term—constructed on a trade-weighted (rather than GDP-weighted) basis—was included in the regression but not found significant.

⁴⁹The effects of unification are introduced through a series of time dummies for 1990–93 in the current account equation for Germany. The dummies have the expected negative sign, reflecting the significant decline in the saving-investment balance (as a share of GDP), reaching a peak decline in 1991 with a point estimate of –3 (percentage points) followed by –1 and –1½ (percentage points) in subsequent years. As described later, in constructing the saving-investment norm, we adjust Germany's country-specific constant to broadly reflect the (persistent) effects of unification, with counterpart adjustments in the saving-investment balances of Germany's partner countries (distributed according to trade shares). We further impose a gradual unwinding of these effects at a 10 percent annual rate of decay over time.